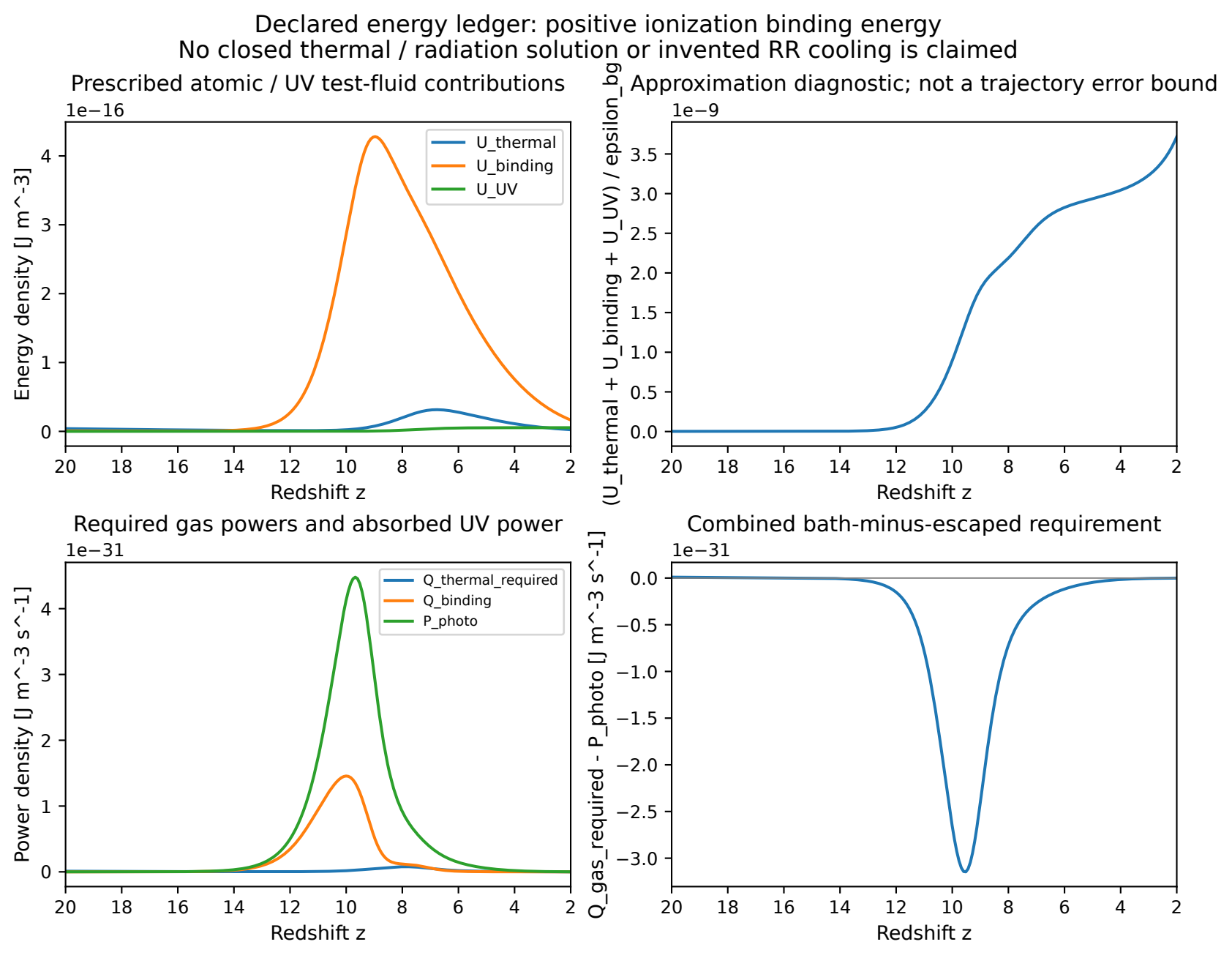
Declared energy ledger: positive ionization binding energy
No closed thermal / radiation solution or invented RR cooling is claimed
Prescribed atomic / UV test-fluid contributions
1e−16
Energy density [J m^-3]
U_thermal
U_binding
U_UV
Redshift z
Approximation diagnostic; not a trajectory error bound
1e−9
(U_thermal + U_binding + U_UV) / epsilon_bg
Redshift z
Required gas powers and absorbed UV power
1e−31
Power density [J m^-3 s^-1]
Q_thermal_required
Q_binding
P_photo
Redshift z
Combined bath-minus-escaped requirement
1e−31
Q_gas_required - P_photo [J m^-3 s^-1]
Redshift z